# Robotics Programming Lab

Research Metric Dictionary v1.1 - V5.2 Three-Phase Plan

55 implemented basic metrics | 161 total candidate metrics

## Purpose and interpretation

This version assigns every candidate metric to a delivery phase and defines the 55 metrics implemented in V5.2. Phase 1 reports directly observable learning activity. Phase 2 adds revision, misconception, survey, and study-condition evidence. Phase 3 adds AI-quality and experimental statistical analysis. Only working Phase 1 metrics appear in the application.

| Phase | Metric IDs | Purpose | Status |
|---|---|---|---|
| Phase 1 - Basic | POP-01, POP-03-08; OUT-01-05, 07-12, 15; ATT-01-03; TIME-01-05, 09-10; ERR-01-08, 16; TUT-01, 02, 04; ASG-01-06; SYS-09-15, 18 | Progress, performance, time, safe errors, basic tutor use, assignment summaries, and data health | Implemented in staging |
| Phase 2 - Moderate | POP-02, 09-10; OUT-13-14; ATT-04-12; TIME-06-08, 11-12; ERR-09-15; LRN-01-08; TUT-03, 05-18; ASM-01-03; SUR-01-04, 06-07; ASG-07, 09, 11-12; SYS-01-08, 16-17 | Revision behavior, recovery, objectives, misconceptions, tutor use, surveys, conditions, and operations | Roadmap only |
| Phase 3 - Advanced | OUT-06; AIQ-01-12; ASM-04-06; SUR-05; ASG-08, 10; CMP-01-15 | AI correctness and safety, experimental comparisons, effects, confidence intervals, and thesis analysis | Roadmap only |

| Rule | V5.2 definition |
|---|---|
| Authoritative evidence | Persisted PostgreSQL submissions determine scores, pass status, execution category, assignment version, and timestamp. |
| Completion | First valid Run or Submit attempt with passed=true. |
| Active time | 30-second heartbeats while the page is visible and the student interacted within the previous 15 minutes. |
| Historical time | Missing pre-V5.2 timing is null with an availability status and displays as Not available, never zero. |
| Version boundary | Every calculation groups by assignment ID and exact assignment version. |
| Privacy | Students see self only. Instructors see participant codes and safe aggregates. Hidden identities/content, login IDs, source code, credentials, and solutions are excluded. |
| AI boundary | AI interaction counts describe optional tutor use and never change deterministic test-case grades. |

# Implemented Phase 1 metric catalog

The following definitions match the authenticated machine-readable catalog returned by GET /research/metrics. All entries have implementation status implemented.

| ID | Metric | Definition | Formula | Unit | Audience |
|---|---|---|---|---|---|
| POP-01 | Assigned students | Active non-demo students assigned to published assignments. | distinct active non-demo student users | students | instructor |
| POP-03 | Students started | Assigned students with meaningful assignment activity or a valid attempt. | distinct started students | students | instructor |
| POP-04 | Participation rate | Share of assigned students who started. | 100 * students_started / assigned_students | percent | instructor |
| POP-05 | Students completed | Started students with at least one successful current-version assignment. | distinct students with passed=true | students | instructor |
| POP-06 | Completion rate | Share of started students who completed at least one assignment. | 100 * students_completed / students_started | percent | instructor |
| POP-07 | Assignment uptake rate | Share of assigned students who started an assignment version. | 100 * assignment_starters / assigned_students | percent | instructor |
| POP-08 | Return-session rate | Share of started students with two or more active sessions. | 100 * students_with_2plus_sessions / students_started | percent | instructor |
| OUT-01 | First-attempt score | Score on the first valid attempt for an assignment version. | first valid attempt score | percent | student, instructor |
| OUT-02 | Final-attempt score | Score on the latest valid attempt for an assignment version. | latest valid attempt score | percent | student, instructor |
| OUT-03 | Best-attempt score | Highest valid score for an assignment version. | max(valid attempt score) | percent | student, instructor |
| OUT-04 | Absolute score gain | Change from first to latest valid score. | latest score - first score | percentage_points | student, instructor |
| OUT-05 | Best score gain | Change from first score to best score. | best score - first score | percentage_points | student, instructor |
| OUT-07 | Successful solution | Whether a valid attempt passed the assignment version. | 1 if any valid passed attempt else 0 | boolean | student, instructor |
| OUT-08 | First-attempt success rate | Share of starters who passed on their first valid attempt. | 100 * first_attempt_successes / students_with_valid_attempt | percent | instructor |
| OUT-09 | Overall success rate | Share of assignment starters who completed it. | 100 * completed_students / started_students | percent | instructor |
| OUT-10 | Test pass percentage | Share of all executed tests passed on the latest valid attempt. | 100 * passed_tests / executed_tests | percent | student, instructor |
| OUT-11 | Public-test pass percentage | Share of public tests passed on the latest valid attempt. | 100 * passed_public_tests / executed_public_tests | percent | student, instructor |
| OUT-12 | Hidden-test pass percentage | Aggregate hidden-test pass share without test identities. | 100 * passed_hidden_tests / executed_hidden_tests | percent | instructor |
| OUT-15 | Assignment completion status | Version-specific state: not started, started, partial, or completed. | state from activity, attempts, and first success | category | student, instructor |
| ATT-01 | Total valid attempts | Student-related attempts excluding infrastructure failures. | count(valid attempts) | attempts | student, instructor |
| ATT-02 | Attempts to first success | Sequence number of the first successful valid attempt. | index(first passed valid attempt) | attempts | student, instructor |
| ATT-03 | Unsuccessful attempts | Valid failures before first success, or all valid attempts for non-completers. | failures before success or deadline | attempts | student, instructor |
| TIME-01 | Active working time | Heartbeat time while visible and recently interactive. | sum(accepted active heartbeat seconds) | seconds | student, instructor |
| TIME-02 | Time to first attempt | Accumulated active time at the first valid attempt. | first_attempt.active_seconds_snapshot | seconds | student, instructor |
| TIME-03 | Time to first success | Accumulated active time at first successful attempt. | first_success.active_seconds_snapshot | seconds | student, instructor |
| TIME-04 | Time to final attempt | Accumulated active time at the latest valid attempt. | latest_attempt.active_seconds_snapshot | seconds | student, instructor |
| TIME-05 | Median time between attempts | Median difference between consecutive active-time snapshots. | median(current_snapshot - previous_snapshot) | seconds | student, instructor |
| TIME-09 | Session count | Count of sessions containing recorded active time. | count(sessions where active_seconds > 0) | sessions | student, instructor |

| ID | Metric | Definition | Formula | Unit | Audience |
|---|---|---|---|---|---|
| TIME-10 | Median session duration | Median active duration across non-empty sessions. | median(session.active_seconds) | seconds | student, instructor |
| ERR-01 | Syntax-error count | Valid attempts classified as student syntax errors. | count(status=syntax_error) | attempts | student, instructor |
| ERR-02 | Runtime-error count | Valid attempts classified as student runtime errors. | count(status=runtime_error) | attempts | student, instructor |
| ERR-03 | Timeout count | Valid attempts terminated for student-code timeout. | count(status=timeout) | attempts | student, instructor |
| ERR-04 | Failed-test count | Failed executed public tests across valid attempts. | sum(public tests where passed=false) | tests | student, instructor |
| ERR-05 | Unsafe-code rejection count | Attempts rejected by the execution safety policy. | count(status=rejected and reason!=size_limit) | attempts | student, instructor |
| ERR-06 | Excessive-output count | Attempts whose captured output reached the configured limit. | count(output_truncated=true) | attempts | student, instructor |
| ERR-07 | Oversized-submission count | Attempts rejected because submitted code exceeded the size limit. | count(rejection_reason=size_limit) | attempts | student, instructor |
| ERR-08 | Unique error categories | Distinct student-caused error categories for an assignment version. | count(distinct student error status) | categories | student, instructor |
| ERR-16 | Error-free attempt rate | Share of valid attempts without a student execution error. | 100 * error_free_attempts / valid_attempts | percent | student, instructor |
| TUT-01 | Built-in hint requests | Successfully delivered non-AI progressive hints. | count(delivered fallback progressive hints) | interactions | student, instructor |
| TUT-02 | AI interaction count | Recorded valid requests for AI tutoring. | count(valid AI tutor requests) | interactions | student, instructor |
| TUT-04 | Hint level reached | Highest delivered progressive hint level. | max(delivered hint_level) | level | student, instructor |
| ASG-01 | Median final score | Median latest valid score among assignment starters. | median(student final scores) | percent | instructor |
| ASG-02 | Score distribution | Counts of final scores in fixed score bands. | count(final scores by 0-59,60-69,70-79,80-89,90-100) | students | instructor |
| ASG-03 | Median attempts to success | Median first-success attempt number among completers. | median(attempts_to_first_success) | attempts | instructor |
| ASG-04 | Median active completion time | Median active seconds to first success among completers with tracked time. | median(active_seconds_to_completion) | seconds | instructor |
| ASG-05 | Most-failed tests | Public tests ranked by distinct affected students. | rank public tests by distinct failing students | students | instructor |
| ASG-06 | Most-common errors | Safe error categories ranked by affected students. | rank categories by students with count > 0 | students | instructor |
| SYS-09 | Missing-event rate | Submission workflows missing their expected research event. | 100 * missing submission events / persisted submissions | percent | instructor |
| SYS-10 | Duplicate-event rate | Repeated logical event groups among recorded events. | 100 * duplicate event groups / events | percent | instructor |
| SYS-11 | Orphan-record count | Research records whose required student parent is absent. | count(orphan submissions and learning events) | records | instructor |
| SYS-12 | Version-missing rate | Attempts without a valid assignment version. | 100 * missing-version attempts / attempts | percent | instructor |
| SYS-13 | Condition-missing rate | Study attempts without a participant condition when conditions apply. | not applicable until Phase 2 condition assignment | percent | instructor |
| SYS-14 | Timestamp-order error rate | Sessions or events with impossible timestamp ordering. | 100 * invalid timestamp chains / checked chains | percent | instructor |
| SYS-15 | Export reconciliation | Whether export row counts match authoritative database counts. | exported matching rows / expected rows | status | instructor |
| SYS-18 | Demo-data contamination count | Persistent research records linked to demo accounts. | count(persistent demo submissions and events) | records | instructor |

Catalog version 1.1. Metric counts: Phase 1 = 55, all phases = 161. Phase 2 and Phase 3 definitions remain in the original v1.0 candidate dictionary and the V5.2 phase roadmap until implementation data contracts are approved.